

Can Personalized Agents Predict Social Experience?

Personalization, learning from encounters, and efficient matching

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Research proposal for discussion with Michael Bernstein · September 2026

1. Research Question

Can a personalized large language model (LLM) predict how compatible someone will feel with a person they have never met, given the connection they seek? An agent representing someone’s social preferences should be evaluated against that person’s experience. Park and colleagues applied this principle to agents grounded in self-reports, assessing predictions relative to participants’ own test–retest consistency.¹

We benchmark *human–LLM alignment*: how closely predictions match post-conversation compatibility ratings, given each participant’s goal. These ratings are directional: A’s rating of B may differ from B’s rating of A. We test initial personalization, improvement through brief interviews alternating with matching and feedback, and score reconstruction through a recommender system. Together, these comparisons ask how accurately agents predict personal experience, how much human input helps, and how many LLM calls are needed.

2. Participant Experience

We propose structured in-person introductions for students arriving at Stanford. Each round involves a previously unfamiliar partner; the pilot will determine round count and conversation duration.

1. **Initial interview.** Before the first encounter, a text interview asks who participants are and what connection they seek. Approximately one minute and five or six questions are tentative pilot targets; their feasibility and adequacy remain to be tested.
2. **Prediction.** All four model conditions record predictions before the conversation. The application coordinates assignments; participants see neither predicted scores nor private agent states.
3. **Conversation.** Partners meet with a common time limit and opening prompt.
4. **Private rating.** Each answers: “How compatible did you feel with this person, given the connection you were seeking?” on a scale from 1 (not at all) to 7 (extremely), reports goal fulfillment and comfort, and briefly explains the rating.
5. **Update.** Between encounters, participants complete one brief follow-up interview for the next round. A fixed procedure updates adaptive agents’ private states using prior ratings, reflections as applicable, and new self-reports; current goals are recorded separately. Subsequent rounds begin at step 2, with updates tested only on previously unmet partners.

3. Representations and Comparisons

Interview answers serve three roles: a participant-approved description H_i for other agents; a private state $P_{i,0}$ holding self-reports, preferences, and available encounter history; and a separate current goal $g_{i,0}$. Keeping goals separate distinguishes the connection sought now from broader preferences.

For example, a student may seek friends at one event and collaborators at another. After a conversation, they might report that feeling listened to mattered more than shared interests. Their agent can use this feedback in predictions for new partners. Before encounter r , a personalized agent predicts

$$O_r(i, j) = f_\theta(H_i, P_{i,r}, g_{i,r}, H_j) \in [1, 7],$$

where H_j describes the partner and $g_{i,r}$ is the current goal. Reverse predictions are made separately. We compare four conditions:

Generic: no participant self-description or encounter history.

Initially personalized: retains its initial personal representation throughout the study.

Adaptive with ratings: updates the initial state with later interviews and prior ratings.

Adaptive with ratings and reflections: receives the same interviews and ratings, plus explanations.

¹Joon Sung Park et al. (2026). *LLM Agents Grounded in Self-Reports Enable General-Purpose Simulation of Individuals*. arXiv:2411.10109v3, June 28, 2026.

4. Human Evaluation

Every condition receives the same current goal, setting, and candidate description. After the pilot, approved descriptions, model weights, prompt templates, and sampling settings remain fixed. Private states stay fixed during querying; predictions and state checkpoints are retained.

We test whether initial personalization improves predictions over the generic LLM on all encounters, and whether adaptive agents improve on the fixed initial agent on identical later encounters. The latter tests the combined interview-and-feedback update process. Comparing the two adaptive variants, which receive identical later interviews, tests reflections’ incremental contribution.

Primary error is mean absolute error between predictions and human ratings, averaged within participant, then across participants. Within-person ranking complements error. To assess elicitation efficiency, learning curves track alignment gains against feedback count and interview time. Paired error reductions with uncertainty intervals excluding zero support the hypotheses; a precise lack of useful gain or deterioration argues against them, while wide intervals remain inconclusive. Analyses account for repeated participants, partners, dyads, and sessions. The pilot informs power analysis, sample size, and preregistration of comparisons, meaningful effect sizes, and missing-data handling.

5. Efficient Estimation and Matching

Scoring every pair in both directions requires many LLM calls. A separate offline experiment tests whether fewer queries preserve agent score estimates and matching value. A recommender system would use low-rank reconstruction, estimating missing scores from shared patterns across participants and partners, to request further scores and support live matching.

With frozen agent snapshots and dense reference scores for every pair in a smaller cohort, the study compares adaptive, random, and non-adaptive querying on held-out score error, query cost, latency, and lost matching value relative to dense matching under identical constraints.

In live matching, directional scores are combined only for assignment. Predefined randomized exploration includes encounters beyond the highest-scored pairs; analyses distinguish exploratory and recommender-selected encounters. Reconstructed scores estimate agent judgments; human ratings evaluate both direct and reconstructed predictions. Better reconstruction cannot establish better human experiences; that requires a randomized comparison of assignment policies.

6. Contributions

Main contribution: a benchmark of human–LLM alignment on directional, goal-specific compatibility ratings. Three supporting contributions address elicitation, query efficiency, and data for further research:

C1. Alignment through iterative brief interviews, matching, and feedback. Test improvement for new partners and measure alignment gains relative to interview time.

C2. Fewer LLM calls through a recommender system. Test whether sparse reconstruction preserves agent scores and matching value with fewer queries.

C3. A well-distributed longitudinal matching dataset. Link interviews, goals, partner descriptions, assignments, predictions, and human ratings and reflections to study how reported preferences and prediction errors evolve. Potential future uses include evaluating whether open models fine-tuned on SocSci210, a dataset of individual responses to social science experiments,² generalize to interpersonal outcomes, and subsequent fine-tuning with separate development and held-out evaluation data. Reuse would be governed by participant consent.

Negative findings would identify limits of the representation, update process, or reconstruction. Generalization concerns new partners in this population; long-term relationships and other populations remain untested.

Participation is voluntary and subject to appropriate research review and consent.

²Akaash Kolluri et al. (2025). *Finetuning LLMs for Human Behavior Prediction in Social Science Experiments*. EMNLP 2025.